

GURU TEGH BAHADUR PUBLIC SCHOOL, DURGAPUR

SESSION : 2026-27

PRE-BOARD EXAMINATION – I | MARKING SCHEME

CLASS : XII

MAXIMUM MARKS : 50

SUBJECT : COMPUTER SCIENCE (083)

TIME ALLOWED : 1 Hr. 45 Min.

SECTION - A (1 × 13 = 13).

- (a) True (b) (A) 14#7 (c) (B) def calc(a, b=5): (d) (B) pickle (e) (B) 'rb+'
(f) True (g) F.seek(20) (h) (C) Microwave (i) (B) TELNET (j) (B) Bus
(k) (B) bits per second (bps) (l) (A) Both A and R are True and R is the correct explanation for A.
(m) (A) Both A and R are True and R is the correct explanation for A.

SECTION - B

- Q.2** (a) A hub broadcasts incoming data to all connected devices. A switch sends data only to the intended device, using its MAC address. (1) (2)
(b) PPP – Point-to-Point Protocol; POP3 – Post Office Protocol version 3. ($\frac{1}{2} + \frac{1}{2}$)
- Q.3** [11, 10, 3] 3 6 (2)
(1 mark for the list, $\frac{1}{2}$ each for 3 and 6. The list is changed in place because it is mutable; val is unchanged because the integer is immutable.)
- Q.4** write() writes a single string to the file, e.g. f.write("Hello\n"). (2)
writelines() writes all the strings in a list or other iterable, e.g.
f.writelines(["A\n", "B\n"]).
(1 mark for the difference, $\frac{1}{2}$ mark for each example)

SECTION - C

- Q.5**

```
def COUNT_CHARS():
```

 (3)
 f = open("DATA.TXT", "r")
 data = f.read()
 u = l = d = 0
 for ch in data:
 if ch.isupper():
 u += 1
 elif ch.islower():
 l += 1
 elif ch.isdigit():
 d += 1
 print("Uppercase:", u, "Lowercase:", l, "Digits:", d)
 f.close()

(Open and read 1, loop and conditions $1\frac{1}{2}$, display $\frac{1}{2}$)

Q.6 (a) Twinkle (1) (3)
(b) 4 Up (1)
(c) twinkle then 15 on the next line ($\frac{1}{2} + \frac{1}{2}$)

Q.7 **(3)**

```
def FREQ(L):
    D = {}
    for i in L:
        if i in D:
            D[i] += 1
        else:
            D[i] = 1
    return D
```

```
print(FREQ([4, 2, 4, 7, 2, 4]))
```

(Function header $\frac{1}{2}$, loop 1, dictionary update 1, return and call $\frac{1}{2}$. Accept `D[i] = D.get(i, 0) + 1` or any other correct logic.)

SECTION - D

Q.8 **(4)**

```
import csv
def AddBook():
    f = open("LIBRARY.CSV", "a", newline='')
    w = csv.writer(f)
    bid = input("Book ID: ")
    title = input("Title: ")
    author = input("Author: ")
    qty = int(input("Qty: "))
    w.writerow([bid, title, author, qty])
    f.close()

def ShowLowStock():
    count = 0
    f = open("LIBRARY.CSV", "r")
    r = csv.reader(f)
    for rec in r:
        if int(rec[3]) < 5:
            print(rec)
            count += 1
    f.close()
    print("Total books with low stock:", count)
```

(a) 2 marks (b) 2 marks. Also accept a header row skipped with `next(r)`.

Q.9 **(4)**

(a) ARPANET (Advanced Research Projects Agency Network) was the first packet-switched network, started in 1969 by the US Department of Defense. NSFNET evolved from it. (1)

(b) Circuit switching sets up a dedicated physical path before transmission, which stays reserved for the whole session. Packet switching has no dedicated path: data is split into packets that may travel by different routes. (1)

(c) Microwaves travel in straight lines and cannot bend around obstacles, so the sending and receiving antennas must be aligned and in direct sight of each other. (1)

(d) A domain name is a human-readable name that identifies a website and maps to its IP address. Domain name here: `www.cbse.gov.in` (also accept `cbse.gov.in`). ($\frac{1}{2} + \frac{1}{2}$)

Q.10 (a)

(4)

10 15
25 40

(1 mark per line)

(b)

Riya-85-A
Amit-40-B
Joy-90-A
Sam-60-B

(½ mark per line)

SECTION - E

Q.11 (a) 'wb' creates a new file, or erases the existing contents, before writing. 'ab' keeps the existing contents and adds new data at the end. (1) **(5)**

```
import pickle
def addrec():
    f = open("STUDENT.DAT", "ab")
    adm = int(input("Admission No: "))
    name = input("Name: ")
    per = float(input("Percent: "))
    pickle.dump([adm, name, per], f)
    f.close()

def update_percent(adm):
    records = []
    found = False
    f = open("STUDENT.DAT", "rb")
    try:
        while True:
            rec = pickle.load(f)
            if rec[0] == adm:
                rec[2] += 5
                found = True
                records.append(rec)
    except EOFError:
        f.close()
    if found:
        f = open("STUDENT.DAT", "wb")
        for rec in records:
            pickle.dump(rec, f)
        f.close()
    else:
        print("Record not found")
```

(b) 2 marks

(c) 2 marks

(c) Also accept an in-place update using 'rb+' with `tell()/seek()`, or writing to a temporary file and renaming it.

- Q.12** (a) TECH block. It has the most computers (120), so placing the server there keeps most traffic local (80–20 rule). (1) (5)
- (b) Star layout centred on TECH: TECH–ADMIN (60 m), TECH–HR (50 m), TECH–ACCOUNTS (110 m).
- Also accept the minimum-cable layout TECH–HR (50 m), TECH–ADMIN (60 m), HR–ACCOUNTS (70 m). (1)
- (c) (i) Repeater: in the star layout, between TECH and ACCOUNTS, because that distance (110 m) is more than 100 m. In the minimum-cable layout no segment is over 100 m, so no repeater is needed. Give credit for either answer if it matches the student's own layout.
- (ii) Switch: in every block, to connect the computers within that block. ($\frac{1}{2} + \frac{1}{2}$)
- (d) WAN (Wide Area Network). (1)
- (e) VoIP (Voice over Internet Protocol). (1)

***** End of Marking Scheme *****